COMPUTER SYSTEMS

Introduction

The word computing comes from the terms INFORMATION and AUTOMATIC, in this way, computing can be defined as *the automatic processing of information through electronic devices and computer systems*.

It is very common to confuse the terms *information* and *computer system*, but we have to differentiate them clearly. An information system is a set of elements oriented to the treatment and administration of data and information, organized and ready for later use, which could be a filing cabinet like the one in the image.



A computer system (Computer System) **is an information system that is computerized, a set of parts or resources, which are related to each other to store and process information with a common goal**. Not all information systems are computer systems, but all computer systems are information systems. Computer systems must have the ability to fulfill three basic tasks input (information capture), processing and output (transmission of results).

The most general purpose or objective of computer systems is to make tasks faster, more flexible and more comfortable for users, using computer technology efficiently for this purpose.

Specifically, the basic objectives of a computer system should be:

- Reduce time, costs and effort in a system.
- Streamline an existing system, which can be manual, or even computerized but old or obsolete.
- Create a new system, to solve a specific problem, perhaps integrating it into an existing system.
- Capture data from your own source.
- Reduce the number of manual tasks, thus reducing the number of possible errors.
- Centralize process control.
- Increase the productivity of a company.

Elements that make up a Computer System

Before analyzing the elements that are part of a Computer System, we have to understand that structurally a Computer System can work even if its elements are separated, but not functionally. If one of the elements that is part of a Computer System is not present, it will not work.

That is why we can understand an IS as a user using a small personal computer, or we can take a company in its entirety with all its employees and computer equipment.



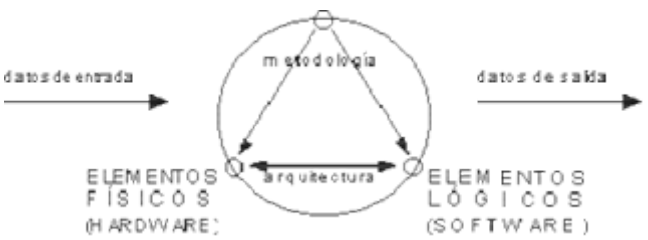


We will take the following definition of Computer System from now on so as not to get lost in the broad sense of the term:

Computer-based information processing system.

The elements of a computer system are as follows:

- Physical elementelement that corresponds to the physical part of the machines. Also called **hardware**, we find it in elements such as the computer as well as in peripherals or communication elements. It provides the processing and calculation capacity in a Computer System.
- Logical elementit is the non-tangible part of a Computer System, it is also said of everything programmable in a Computer System. It is every element that cannot be touched, instructions and orders that make up the programs. It can be written in any programming language. It is also known as **software** and its objective is to take advantage of the processing capacity of the hardware to carry out its operations on the information entered to process it (modify it, save it, ...) and be able to extract it in some way.
- Human elementit is the element that people form. Any person or group of people involved in the design, operation and administration of an IS
- Information elementAlthough sometimes it is not taken as part of a Computer System, it is all that information that helps the human element to use the physical and logical elements.



Therefore we can say that a Computer System is used to process (save, modify, ...) some data that is entered externally, to show the result of that processing in some way, controlled by the human element.

For this reason, it is often said that the human element is not part of a computer system, but we know that it is part of it, since without the interaction of the human element an IS would not fulfill any function.

This data processing, as we have seen, is done through the logical element and the physical element of an IS and the first thing that comes to mind when we talk about an IS is a computer. A computer *is a set of hardware and software that allows us to process data* and see the result on a screen (normally).



At the beginning, computers (or computers) were machines that worked thanks to mechanical elements and their processing capacity was not even comparable to those of today. In the 19th century, with the appearance of electromechanical components, their performance increased, and with the appearance of transistors and integrated circuits, enormous advances in processing were achieved.

Also due to the initial costs of computers, at first they were only accessible for a few (mainly military uses and research), but with the appearance of cheaper elements they were introduced in other areas until today in which practically anyone can have a computer at home for daily use.

Classification of computer systems

As we have seen when we have defined what a Computer System is, we have seen that it has a broad meaning since it can be considered the sum of several elements working together. Depending on what the purpose of that work is, we can classify Computer System into different groups:

We will make the first classification based on the use of that specific Computer System:

- **Computer systems for general use.** They are systems that can be used for more than one specific purpose (this is the case of personal computers).
- **Computer systems for specific use**. They are those that fulfill a single specific use, they are designed to fulfill only one purpose. They use one or several programs and normally the physical element is also adapted to fulfill a single use, an example can be the robots used in the industry (milling machines, presses, welding robots,..... .....)

Another classification that we can make is when observing the benefits of the Computer System:

- **Supercomputers**they are very powerful computers (even highly connected to each other), with many CPUs working together. They usually have scientific uses (astronomy, genetics,...) or research and, of course, also military applications.
- Large Systems, Central Computers or **Mainframes**These are equipment used to support large communications networks with hundreds or even thousands of users. For example, for the processing of bank transactions.
- **Servers**They are normally responsible for providing services through a network. They are computers with great input and output capacity. They are widely used on the Internet and in Local Networks.
- **Workstations**These are very powerful and specialized single-user computers that allow for increased performance (for example, for design and CAD). Workstation hardware is being optimized for situations that require high performance and reliability, where a microcomputer might stop responding.
- **Microcomputers**These are less powerful computers that can be classified in various ways. These includeprofessional and personal computers, office and home computers, and desktop and laptop computers.
- **Pocket Devices.** They are devices that fit in the hand and perform many functions. The current trend is to reduce devices more and more , an example is the PDA (Personal Digital Assistant) that performs functions of managing contact agendas, writing notes,... or today's smartphones.

It is important to note that the performance of a system depends more on the relationship between its components and how they are combined with each other than on the operation of each of them individually. For this reason, the operation of an Information System is not only improved by combining the best physical, logical and personnel systems, but also by the harmonization and coordinated operation of its elements.

In general, in any system it is difficult to assess the general behavior and its performance, due to the fact that many variables intervene, and there is no single parameter that indicates the result of its operation. To evaluate the performance of a computer system, more or less complex techniques are used, such as statistical methods or queuing theory.

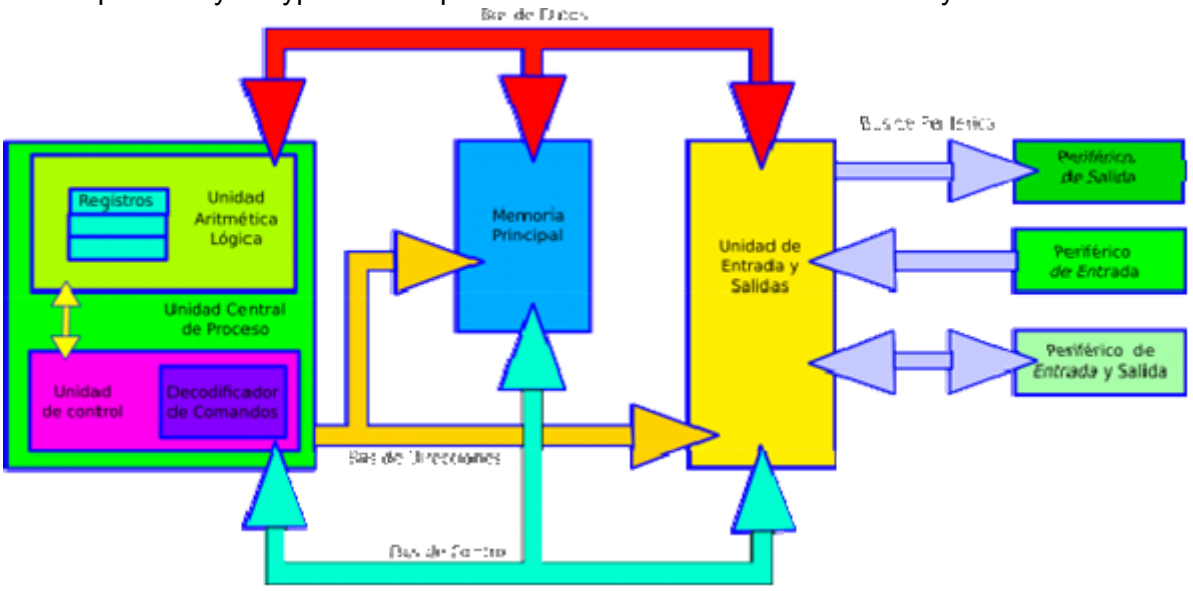
Evaluation techniques are applied to performance indices such as:

- **Productivity**amount of work done by the system in the unit of time.
- **Presence time**from the moment a program enters the system and the moment its execution ends.
- **Response time**in interactive systemsfrom when a user makes a request to the system until the response appears.
- **Capacity of a resource**maximum theoretical value.
- **Availability**of the system or of a resource.
- **CPU utilization rate**

MIPS (millions of instructions per second) or **Megaflop** (millions of floating per second) are used to measure the power of computers of the same family. To compare teams of different nature, it is more correct to use standardized test programs.

Physical element. Hardware

The layout of modern computers follows the design of the mathematician and physicist John Von Neumann (1903-1957). It was a design in which 4 basic units are differentiated that work interconnected to process the necessary information. It is a fairly old design but it is still used in practically all types of computers and devices that we use in today's life.



The 4 units in which the design is divided are the following:

- **CPU** (Central Processing Unit). the main processing unit is responsible for processing the information. in turn are divided into 2 other units.
- Arithmetic-logical unit (**ALU**). This unit is responsible for carrying out logical and arithmetic processes, which is essentially the processing of information.
- Control unit (**UC**). It is the unit that controls the signals that are sent to the other elements and ensures that they all work well and that the instructions are executed correctly.
- **Memory unit**. It is the unit that is used to store the instructions of the programs as well as the data that these programs use. It is made up of memory cells that are capable of storing the smallest memory unit (the bit).
- **unit** . **O**/Unit connected to the other units (memory and CPU) and in charge of introducing input information from the outside and once processed to get it out again.

Although it is not considered a unit itself, we also have to take into account that all these units are interconnected with each other by means of buses. Buses **.** are channels through which information flows uni or bidirectional depending on the bus

This unit design does not correspond to the parts of a computer, since it is a theoretical design. In practice we have elements such as processors, hard drives, USB sticks, graphics cards,... which would be elements that each correspond to a Von Neumann design unit.

Logic element. Software

Computer software is a collection of stored data-related computer programs that tell the computer what to do and how to do it. In other words, software is a set of programs, procedures, algorithms and their documentation involved in the operation of a data

processing system. A piece of software performs the function of the program it implements, either by directly providing instructions to the computer's hardware or by serving as input to another piece of software. In contrast to hardware, software "can't be touched."

Software is also sometimes used in a more defined sense, referring only to a certain application. Although this distinction is somewhat arbitrary, and sometimes confusing, software can be classified as follows:

- **System software.** It is the one that allows the hardware to work. Its objective is to adequately separate the programmer from the details of the particular computer being used, isolating him especially from the processing related to the internal characteristics ofmemory, disks, ports and communications devices, printers, screens, keyboards, etc. The system software provides the user and programmer with adequate high-level interfaces and support utilities that allow its maintenance. It includes among others:
  - operating systems
  - Device drivers
  - Diagnostic tools
  - Correction and Optimization Tools

- Programming software. It is the set of tools that allow the programmer to develop computer programs, using different alternatives and programming languages, in a practical way. It includes among others:
  - Text editors
  - Compilers
  - Interpreters
  - Linkers
  - Scrubbers
  - Integrated Development Environments (IDE)

- **Application software,** allows users to carry out one or more specific tasks, in any field of activity that can be automated or assisted, with special emphasis on business. It includes among others:

  - Industrial automation and control applications
  - Office applications
  - Educational software
  - Medical software
  - Numerical Calculation Software
  - Aided Design Software (CAD)
  - Numerical Control Software (CAM)

Types of software

Since the technology companies started making the use of their software more restrictive in the 1980s, with the computer boom, the number of existing licenses has multiplied exponentially. These are **licenses** that are, basically, a contract between the author of the program and the user in which a series of terms and clauses are established that the user must comply with in order to use the software.

It is very important to differentiate between **proprietary** and **free software**. The first concept applies to any computer program whose use, redistribution or modification is prohibited or requires the express permission of the owner of the software.

For its part, **free software** refers to the computer program in which the user has **freedom** over it, and can be **used**, **copied**, **modified** and **distributed** as the user wants. Its potential is great, since it allows millions of programmers to apply and expand its growth, and the fact is that the availability and global access to this type of software allows the expansion of the knowledge deposited in each piece developed.

Some of its most important advantages are that it is cheaper than proprietary software and, the fact that you can access its source code, offers a certain technological independence, allowing you to develop your own version of the software.

Next we are going to clarify certain terms that usually create confusion between them:

- Free SoftwareFree software is software that is provided with permission for anyone to use, copy, and/or distribute it, either with or without modification, for free or for a fee. In particular, this means that the source code must be available. "If it's not source, it's not software"
- Open Source SoftwareSome people use the term "open source" software to refer to more or less the same category as free software. However, they are not exactly the same type of softwaresome licenses are accepted that are considered too restrictive, and there are free software licenses that are not accepted as open source. However, the differences between what the two categories encompass are fewalmost all free software is open source, and almost all open source software is free.
- Proprietary softwareProprietary software is another name for software that is not free. In the past we had subdivided non-free software into "semi-free software", which could be modified and redistributed without commercial purposes, and "proprietary software", which could not be modified or redistributed. But we have abandoned this distinction and now use the term "proprietary software" as a synonym for non-free software.
- Public domain software is software that is not copyrighted. If the source code is in the public domain, it is a special case of unlicensed free software, which means that some copies or modified versions may not be free at all. In some cases, an executable program may be in the public domain but the source code may not be freely available. In that case it is not free software, because free software requires accessibility to the source code. On the other hand, most free software is not in the public domain but under copyright, and the copyright holders have given legal permission for everyone to use it freely, using a free software license.

Depending on the way in which we want to distribute software, we have different licenses available that will guarantee that the use, redistribution or ability to sell it is one way or another. Let's see what are the main ones today:

- CopyrightThis is the best known license, and unfortunately the most widely used. Normally the only person who can do what they please with your work is the author of it. In other words, the work has an owner and if you want it, you have to pay for it in addition to asking for permission to use it. But let's take the case that you obtain the work of this author or company, because you will NOT be able to legally redistribute it, because the only one qualified to do so is the author of the work, that is, this license only allows you to have the work for personal use. This is applied among othersin record companies, books, plays, movies and software.
- Copyleftis free software whose distribution terms guarantee that all copies of all versions have approximately the same distribution terms. This means, for example, that copyleft licenses generally do not allow third parties to add additional requirements to the software (although they may be allowed to add a limited set of requirements that are considered safe) and require that the source code be made available. This protects the program and its modified versions against some of the most common ways of turning it into proprietary software.

Within this type of licenses others have been born, which are the ones that are currently used, such asCreative Commons, GPL, BSD, ... All these licenses ensure distribution, modification and profit in such a way that each one of them requires certain clauses that must be fulfilled.

On the other hand, in proprietary software, there are also other types of licenses that mark in some way the basic aspects of the use of software, such as:

- FreewareThe term *freeware* does not have a clearly accepted definition, but is generally used to refer to packages in which redistribution is allowed but modification is not (and whose source code is not available). These packages **not** free software, so we urge you not to use the term "freeware" to refer to free software.
- SharewareThe term shareware refers to software of which copies are allowed to be redistributed, but whoever continues to use a copy must pay to obtain the license. Shareware software is not free software, or even semi-free software, for two reasons:

For most shareware programs, the source code is not available, so they cannot be modified.

Shareware software does not come with permission to make a copy and install it without paying the license, even for people involved in non-profit activities. In practice, users often ignore the distribution terms and do so anyway, even though the conditions do not allow it.





